

- 5 The curve C has polar equation $r^2 = \frac{1}{\theta^2 + 1}$, for $0 \leq \theta \leq \pi$.

(a) Sketch C and state the polar coordinates of the point of C furthest from the pole. [3]

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(b) Find the area of the region enclosed by C , the initial line, and the half-line $\theta = \pi$. [4]

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and verify that this equation has a root between 1.1 and 1.2.

[5]

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